

Practical 3 — Functional diversity indices and null models

Master FBE · Functional Ecology in R

Using the community matrix and trait data from practicals 1 and 2, you will compute and compare functional diversity across biogeographic realms, test for functional structure with null models, and examine the relationship between functional and taxonomic diversity.

```
library(FD); library(TPD); library(ggplot2); library(dplyr)

comm  <- read.csv("fish_communities.csv", row.names=1)
traits <- read.csv("fish_traits.csv", row.names=1)
env    <- read.csv("env_data.csv", row.names=1)
```

Question 1 — Community-weighted mean (CWM) (15 pts)

1a. Compute CWM for all traits across basins. Map body length CWM on a world map — do warmer regions host larger-bodied fish?

```
# Align species
sp_shared <- intersect(colnames(comm), rownames(traits))
comm_s    <- comm[, sp_shared]
tr_s      <- traits[sp_shared,]

cwm <- funtcomp(tr_s, as.matrix(comm_s))
cwm$Realm <- env[rownames(cwm), "Realm"]

# Body length CWM by realm
ggplot(cwm, aes(reorder(Realm, BL_mean), BL_mean, fill=Realm)) +
  geom_boxplot() + coord_flip() + labs(y="CWM body length (cm)") + theme_bw()
```

1b. Test whether water temperature (env\$temp_mean) predicts CWM body length. Interpret the slope in an ecological context (Bergmann's rule in fish?).

```
cwm$temp <- env[rownames(cwm), "temp_mean"]
m <- lm(BL_mean ~ temp, data=cwm)
summary(m)
plot(cwm$temp, cwm$BL_mean, pch=16, col="#e67e22",
      xlab="Mean temp (°C)", ylab="CWM body length")
abline(m, lwd=2)
```

Question 2 — Functional diversity indices (25 pts)

2a. Compute all FD indices (FRic, FEve, FDiv, FDis, RaoQ) using dbFD().

```
set.seed(42)
fd <- dbFD(x=tr_s, a=as.matrix(comm_s), m=3,
           calc.FRic=TRUE, calc.CWM=FALSE, print.pco=FALSE)

fd_df <- data.frame(
```

```

basin=rownames(comm_s),
S=specnumber(comm_s),
FRic=fd$FRic, FEve=fd$FEve,
FDiv=fd$FDiv, FDis=fd$FDis, RaoQ=fd$RaoQ,
Realm=env[rownames(comm_s), "Realm"]
)

```

2b. Is FRic correlated with species richness? This is a known confound — plot the relationship and compute Spearman r.

```

cor.test(fd_df$S, fd_df$FRic, method="spearman")
ggplot(fd_df, aes(S, FRic, colour=Realm))+
  geom_point(size=2.5, alpha=0.8)+
  geom_smooth(method="lm", colour="black", linetype=2)+
  labs(x="Species richness S", y="FRic")+theme_bw()

```

2c. Compare FRic across realms with a boxplot. Which realm is functionally richest? Does it match the realm with highest species richness?

```

ggplot(fd_df, aes(reorder(Realm, FRic, median), FRic, fill=Realm))+
  geom_boxplot(alpha=0.7)+coord_flip()+
  labs(x="Realm", y="FRic")+theme_bw()+theme(legend.position="none")

```

Question 3 — Null models and SES (25 pts)

3a. Compute the standardised effect size (SES-FRiC) for each basin using a species-shuffling null model (nperm=199 for speed).

```
set.seed(42)
null_FRic <- replicate(199,{
  tr_null <- tr_s[sample(nrow(tr_s)),,drop=FALSE]
  rownames(tr_null) <- rownames(tr_s)
  tryCatch(
    dbFD(tr_null,as.matrix(comm_s),m=3,
      print.pco=FALSE,calc.CWM=FALSE)$FRic,
    error=function(e) rep(NA,nrow(comm_s))
  )
})

FRic_obs <- fd_df$FRic
SES_FRic <- (FRic_obs-rowMeans(null_FRic,na.rm=TRUE))/
  apply(null_FRic,1,sd,na.rm=TRUE)
fd_df$SES_FRic <- SES_FRic
fd_df$pattern <- ifelse(SES_FRic>1.96,"overdispersed",
  ifelse(SES_FRic< -1.96,"clustered","random"))
```

3b. Plot SES-FRiC by realm. Are Neotropical assemblages functionally overdispersed or clustered? Propose a mechanism.

```
ggplot(fd_df,aes(Realm,SES_FRic,fill=Realm))+
  geom_boxplot(alpha=0.7)+
  geom_hline(yintercept=c(-1.96,1.96),linetype=2,colour="red")+
  geom_hline(yintercept=0,linetype=1)+
  coord_flip()+theme_bw()+theme(legend.position="none")
```

3c. Is SES-FRiC correlated with water temperature? Interpret: does environmental filtering increase or decrease with temperature?

```
fd_df$temp <- env[rownames(comm_s),"temp_mean"]
ggplot(fd_df,aes(temp,SES_FRic,colour=Realm))+
  geom_point(size=2.5)+geom_smooth(method="lm",colour="black")+
  geom_hline(yintercept=0,linetype=2)+
  labs(x="Mean temperature (°C)",y="SES-FRiC")+theme_bw()
```
